IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

ELECTROMAGNETISM

Friday 12 May 14:30

Time allowed: 2 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Formula sheet

Universal constants: $\varepsilon_0 = 8.84 \times 10^{-12}$ F/m $\quad \mu_0 = 4\pi \times 10^{-7}$ F/m $\quad c = 3 \times 10^8$ m/s

Maxwell's equations in integral form

$$\oiint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s} = \iiint_V \rho \mathrm{d}v$$
$$\oiint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s} = 0$$
$$\oint_L \mathbf{H} \cdot \mathrm{d}\mathbf{l} = \iint_S \mathbf{J} \cdot \mathrm{d}\mathbf{s} + \frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s}$$
$$\oint_L \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s}$$

Maxwell's equations in differential form

$$\nabla \cdot \mathbf{D} = \rho$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Constitutive relations: $\mathbf{D} = \varepsilon_0 \varepsilon_\mathrm{d} \mathbf{E} \qquad \mathbf{B} = \mu_0 \mu_\mathrm{r} \mathbf{H} \qquad \mathbf{J} = \sigma \mathbf{E}$

Boundary conditions

$$(\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{n} = \sigma \qquad (\mathbf{B}_1 - \mathbf{B}_2) \cdot \mathbf{n} = 0$$
$$\mathbf{n} \times [\mathbf{E}_1 - \mathbf{E}_2] = 0 \qquad \mathbf{n} \times [\mathbf{H}_1 - \mathbf{H}_2] = \mathbf{J}_\mathrm{s}$$

Solution of Poisson's equation for electrostatic potential

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \iiint_V \frac{\rho(\mathbf{r}')\mathrm{d}v}{|\mathbf{r} - \mathbf{r}'|}$$

Biot-Savart law

$$\mathbf{B} = \frac{\mu_0}{4\pi} I \int_L \frac{\mathrm{d}\mathbf{l} \times \mathbf{r}}{r^3}$$

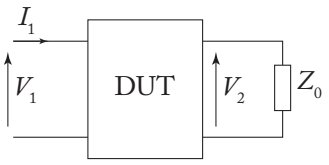
Transmission-line equations

$$\frac{\partial I(z,t)}{\partial z} = -C \frac{\partial V(z,t)}{\partial t} \qquad \text{and} \qquad \frac{\partial V(z,t)}{\partial z} = -L \frac{\partial I(z,t)}{\partial t}$$

Input impedance of a loaded transmission line

$$Z_{\text{in}} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan(kd)}{Z_0 + jZ_{\text{L}} \tan(kd)}$$

S-parameters for a two-port device



$$S_{11} = \frac{V_1 - Z_0 I_1}{V_1 + Z_0 I_1} \quad \text{and} \quad S_{21} = \frac{2V_2}{V_1 + Z_0 I_1}$$

Fields of a short dipole antenna (far-field region)

$$E_\theta = jZ_0 \frac{I_0 kd}{4\pi} \frac{e^{-jkr}}{r} \sin\theta$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Fields of a half-wave dipole antenna (far-field region)

$$E_\theta = \frac{jZ_0 I_0}{2\pi} \frac{\cos\dfrac{\pi\cos\theta}{2}}{\sin\theta} \frac{e^{-jkr}}{r}$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Gauss' and Stokes' theorems

$$\oiint_S \mathbf{F} \cdot \mathrm{d}\mathbf{s} = \iiint_V \nabla \cdot \mathbf{F}\, \mathrm{d}v \qquad \oint_L \mathbf{F} \cdot \mathrm{d}\mathbf{l} = \iint_S [\nabla \times \mathbf{F}] \cdot \mathrm{d}\mathbf{s}$$

Useful integrals

$$\int \frac{\mathrm{d}x}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{a^2 + x^2}) \quad \int \frac{\mathrm{d}x}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2\sqrt{x^2 + a^2}} \quad \int \frac{x\,\mathrm{d}x}{(x^2 + a^2)^{3/2}} = -\frac{1}{\sqrt{x^2 + a^2}}$$

ELECTROMAGNETISM

1. a) A three-dimensional static electric field has a strength of the form

$$\mathbf{E} = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z$$

i) Prove that the field is conservative. Show your working. [5]

ii) Find the electrostatic potential. [5]

b) Maxwell's equations contain charge density, ρ, and current density, $\mathbf{J}$. Give one example of each of the following:

i) A system for which $\rho \neq 0$ and $\mathbf{J} = 0$ [1]

ii) A system for which $\rho = 0$ and $\mathbf{J} \neq 0$ [2]

iii) A system for which $\rho \neq 0$ and $\mathbf{J} \neq 0$ [2]

c) State the EM boundary conditions for a boundary between air and a perfect electric conductor. [5]

d) Two identical loops with a self-inductance L are coupled magnetically to each other. Show that the maximum achievable value of the mutual inductance M between the loops is $M = L$. [8]

e) A one-port device is known to be passive and lossless. Derive an expression for the device S-parameter. [7]

f) Bluetooth is a wireless communication standard that operates at 2.45 GHz. A Bluetooth-enabled printer communicates with a computer located 0.3 metres away. Using calculations, explain whether communication between the printer and the computer takes place in the far-field region of their antennas, or not. [5]

2. a) A uniform dielectric is placed in an external static electric field. Briefly answer the following:

i) How can the atoms in the dielectric be modelled? [5]

ii) What is the difference between bound and free charges? [5]

iii) Why are two electric-field vectors, **E** and **D**, introduced in Maxwell's equations? [5]

iv) What is the volume charge density inside the dielectric equal to? [5]

b) Derive an expression for the elecric field strength of a point charge q embedded in a dielectric with a relative permittivity ε_{d}. [10]

3. A two-port device contains a short-circuited transmission line as shown in Fig. 3.1. The transmission line has a length d and a characteristic impedance Z_0. Assume that the port impedances are equal to Z_0.

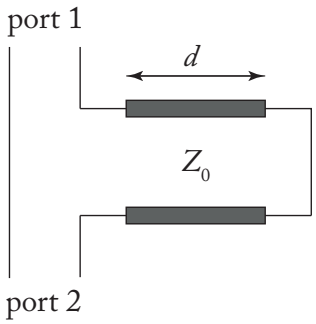


Figure 3.1

a) Find the device S_{11} and S_{21} at the frequency f_0 for which $d = \lambda/4$, where λ is the wavelength of the waves supported by the transmission line. [10]

b) Explain what function the device performs for frequencies around f_0. [5]

c) Explain the simplest way to change the device so that it performs the opposite function. [5]

d) Assume a frequency $f = f_0 + \delta f$, where δf is small ($|\delta f| \ll f_0$). Derive approximate expressions for S_{11} and S_{21} that take into account the smallness of δf. (Hint: $\cot x \approx 1/x$ when $x \approx 0$.) [10]